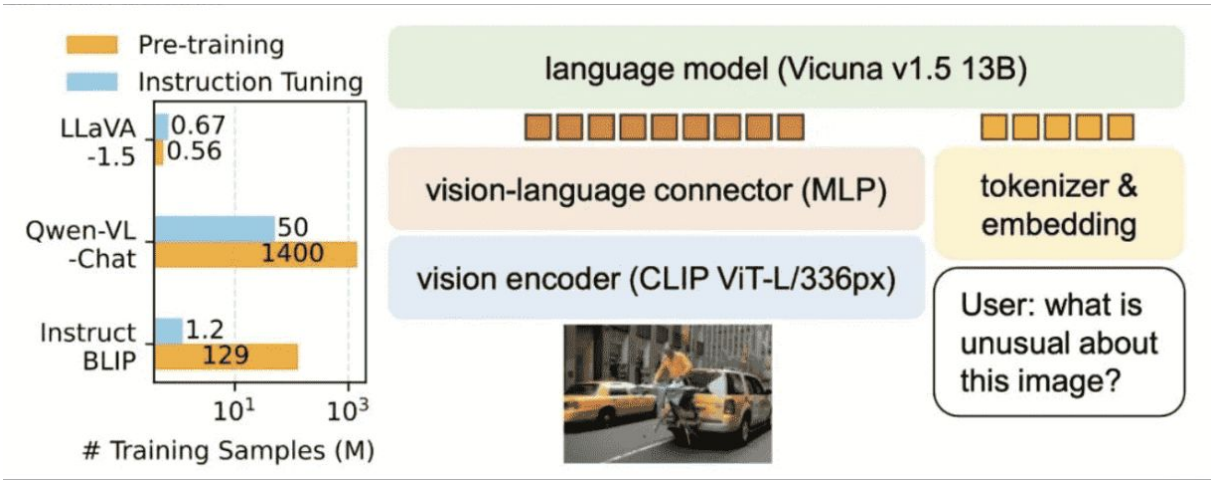
LLaVA-1.5(2023, Microsoft Research)

论文：《Improved Baselines with Visual Instruction Tuning》
论文地址：https://arxiv.org/pdf/2310.03744
项目地址：https://github.com/haotian-liu/LLaVA

相比LLaVA(2023, Microsoft)主要做了几点优化：

- 使用单一的响应格式提示语，明确指示输出格式。例如，“Answer the question using a single word or phrase.”这样的提示语可以促使 LLM 根据用户的指示正确调整输出格式。
- vision-language 的 connector 使用一个两层的 MLP
- 额外使用一些学术任务导向的数据集
- 扩大 LLM 的参数规模、图像的分辨率等

1. 模型架构



1.1 Visual Encoder

CLIP-ViT-L-14 (224^2)

CLIP-ViT-L-336px (336^2)

使用了 Dynamic High-Resolution 策略，用于保留高分辨率图像的细节：

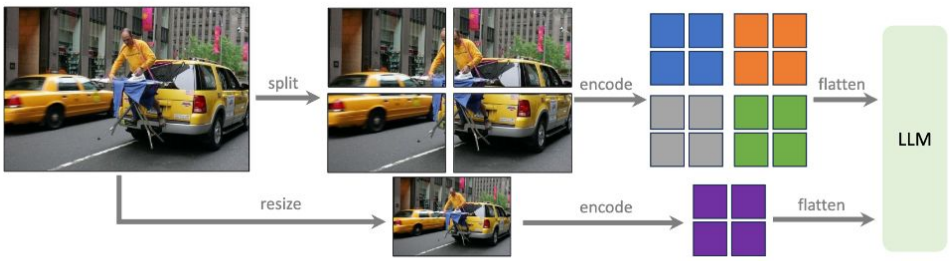


Figure 2. **LLaVA-1.5-HD.** Scaling LLaVA-1.5 to higher resolutions by splitting the image into grids and encoding them independently. This allows the model to scale to any resolution, without performing positional embedding interpolation for ViTs. We additionally concatenate the feature of a downsampled image to provide the LLM with a global context.

注意一个细节，就是 LLaVA 使用的 ViT 特征和BLIP-2(2023, Salesforce Research)一样，是倒数第二层。这可能因为 CLIP 的最后一层特征可能比它之前的层更关注全局和抽象的图像属性，而它之前的层可以更关注有助于理解特定图像细节的局部属性

这个细节可以参考LLaVA(2023, Microsoft)论文：

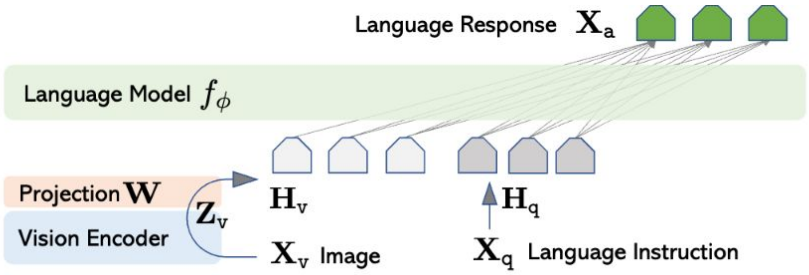


Figure 1: LLaVA network architecture.

For an input image $\mathbf{X}_v$, we consider the pre-trained CLIP visual encoder ViT-L/14 [40], which provides the visual feature $\mathbf{Z}_v = g(\mathbf{X}_v)$. The grid features before and after the last Transformer layer are considered in our experiments. We consider a simple linear layer to connect image features into the word embedding space. Specifically, we apply a trainable projection matrix $\mathbf{W}$ to convert $\mathbf{Z}_v$ into language embedding tokens $\mathbf{H}_v$, which have the same dimensionality as the word embedding space in the language model:

$$\mathbf{H}_v = \mathbf{W} \cdot \mathbf{Z}_v, \quad \text{with} \quad \mathbf{Z}_v = g(\mathbf{X}_v) \tag{1}$$

> 论文中的 grid features 就是 features 的意思，只是说 ViT 会分割成 patch，因此叫“grid features”

并且LLaVA(2023, Microsoft)还做了一个消融实验证明倒数第二层会更好：

Visual features	Before	Last
Best variant	90.92	89.96 (-0.96)
Predict answer first	-	89.77 (-1.15)
Training from scratch	85.81 (-5.11)	-
7B model size	89.84 (-1.08)	-

Table 8: Design choice ablations (%). The difference with the best variant is reported in red text.

1.2 Adapter

two-layer MLP，并且实验证明会比 a linear projection 表现要好

1.3 LLM

LLM 语言模型升级为 Vicuna v1.5 13B，语言模型参数量更大，效果更好

消融实验还测试了以下模型：

LLaMA-based (Vicuna-v1.1, Vicuna-v1.3)

LLaMA2-based (Vicuna-v1.5, LLaMA-2-Chat

2. 模型训练

数据上使用了InstructBLIP(2023, Salesforce Research)使用的：

- VQA (OKVQA, A-OKVQA)
- OCR (OCRVQA, TextCaps)

以及 region-level VQA datasets

- Visual Genome
- RefCOCO

训练模式参考LLaVA(2023, Microsoft)两阶段训练。

模型在三个维度上 scaling 的效果：

	Method	LLM	Res.	GQA	MME	MM-Vet
	InstructBLIP	14B	224	49.5	1212.8	25.6
	Only using a subset of InstructBLIP training data					
0	**LLaVA**	7B	224	–	809.6	25.5
1	+VQA-v2	7B	224	47.0	1197.0	27.7
2	+Format prompt	7B	224	46.8	1323.8	26.3
3	+MLP VL connector	7B	224	47.3	1355.2	27.8
4	+OKVQA/OCR	7B	224	50.0	1377.6	29.6
	Additional scaling					
5	+Region-level VQA	7B	224	50.3	1426.5	30.8
6	+Scale up resolution	7B	336	51.4	1450	30.3
7	+GQA	7B	336	62.0*	1469.2	30.7
8	+ShareGPT	7B	336	62.0*	1510.7	31.1
9	+Scale up LLM	13B	336	**63.3***	**1531.3**	**36.1**

Table 2. **Scaling results** on data, model, and resolution. We choose to conduct experiments on GQA [21], MME [17], and MM-Vet [55] to examine the representative capabilities of VQA with short answers, VQA with output formatting, and natural visual conversations, respectively. *Training images of GQA were observed during training.